

Recreational ecosystem services of mountain lakes in the European Alps: Preferences, visitor groups and management implications

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ABSTRACT

Lakes provide many opportunities for outdoor recreation and tourism, but growing visitor numbers and global changes increase negative effects on recreational values and environmental impacts requiring targeted management strategies. Since there is a lack of information about visitors and their preferences related to natural mountain lakes, this study aimed at assessing people's preferences with a focus on recreational activities, accessibility, and touristic facilities. We collected 526 responses through an online survey and identified recreational visitor groups using cluster analysis. Our results indicate that mountain lakes are mostly valued for passive recreation such as nature observation and relaxation, while recreational activities such as boating or having a barbecue were less often associated with a visit to mountain lakes. Respondents mostly valued hiking/walking trails along the lakeshore as well as direct access to the lake, while they indicated less often the importance of playgrounds and water sports facilities. We identified two major visitor groups: (1) nature-oriented visitors, preferring pristine and remote mountain lakes, and (2) leisure-oriented visitors, favouring mountain lakes that were easily accessible and developed for tourism. Accordingly, the latter group valued most recreational activities at lakes such as reading/listening to music or having a picnic higher. Although there is potential for conflicts between these two visitor groups, the presence of lakes with different level of accessibility may defuse such conflicts. Our findings provide valuable insights for developing targeted management strategies based on the preferences of different visitor groups.

Management implications: Mountain lakes are associated with high recreational values. Maintaining marked hiking trails is essential to provide access, but the construction of tourist facilities needs to be cautiously evaluated, as this can cause a change in the type of visitors with possible negative consequences. The results of the survey also indicate diverging preferences between two major visitor types. Improved and easily accessible tourist facilities at mountain lakes support more recreational uses and may attract leisure-oriented visitors. In contrast, recreational values of nature-oriented visitors could be maintained by preserving the naturalness of remote lakes and not improving accessibility.

1. Introduction

Natural and artificial lakes are highly important for outdoor recreation and tourism, as they provide various opportunities for recreational activities, aesthetic experiences, education and inspiration among others (Allan et al., 2015; Angradi et al., 2016; Fu et al., 2018; Hossu et al., 2019; Reynaud and Lanzanova, 2017; Sterner et al., 2020). Lakes are also often used for promoting tourism destinations (Cagnina et al., 2019; Tuohino, 2006). While large and low elevation lakes are highly

appreciated for water-based activities such as swimming and boating (Allan et al., 2015), natural mountain lakes are more valued for their beauty and are mostly associated with hiking activities (Schirpke, Tasser, et al., 2021; Senetra et al., 2020). Moreover, lakes are increasingly recognized for their positive influence on well-being, providing physical and mental health benefits, as well as generating identity, belonging or inspiration (McDougall et al., 2020; Völker and Kistemann, 2011). Thus, lakes are valued destinations by residents and tourists and, in the light of a globally increasing demand for outdoor recreation

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(Buckley et al., 2015) as well as an intensification of hiking tourism in cooler regions such as the European Alps due to increasing temperatures (Pröbstl-Haider et al., 2020), it can be expected that visitation rates to mountain lakes will further increase.

However, mountain lakes are highly susceptible to global changes (Schmeller et al., 2018), and related recreational values may be affected in several ways. One major factor are higher visitation rates, resulting from an increasing demand for outdoor recreation as well as improved accessibility due to the construction of touristic infrastructure (Dynowski et al., 2019; Schirpke et al., 2019). Moreover, mountain destinations are likely to become more popular with increasing temperatures, thus, replace bathing holidays by hiking holidays in the European Alps (Fleischhacker et al., 2009). Congestion and perceived crowding are strongly associated with higher visitation rates and can negatively influence perceived benefits (Confer et al., 2005; Kainzinger et al., 2015; Roux et al., 2020; Scolozzi et al., 2015; Zehrer and Raich, 2016). Further related threats include higher levels of pollution, noisiness and traffic congestions (Beza, 2010; Confer et al., 2005). Higher visitation rates can also cause conflicts among different visitor groups with different priorities in terms of interests and values (Ament et al., 2017; Schirpke et al., 2020; Scolozzi et al., 2015). For example, people looking for quietness may feel bothered by people carrying out noisy activities (Scolozzi et al., 2015) or anglers may feel disturbed from encountering rafters or kayakers (Confer et al., 2005).

Another reason for possibly decreasing recreational values of mountain lakes is related to the water quality, as aesthetic and recreational benefits are strongly linked to high ecosystem quality (Angradi et al., 2018; Janssen et al., 2020; Keeler et al., 2015; Pröbstl-Haider et al., 2020; Schirpke, Scolozzi, & Tappeiner, 2021a). The water quality of lakes may be affected by increasing levels of eutrophication due to more atmospheric and anthropogenic nutrient inputs combined with higher temperatures (Moser et al., 2019; Roşca et al., 2020; Sadro et al., 2019; Schmeller et al., 2018; Senetra et al., 2020; Van Colen et al., 2018), and thus, lakes can become less suitable for bathing and other recreational ecosystem services (Janssen et al., 2020; Pröbstl-Haider et al., 2020). The introduction of fish for recreational angling purposes in mountain lakes that were naturally fishless has also huge impacts on the entire ecosystem and especially on native amphibian species (Miró et al., 2020), which reduces, for example, possibilities of nature observation (Mocior and Kruse, 2016), while fish feeding negatively influences water quality (Weckström et al., 2016). Additionally, recreational activities in proximity to mountain lakes such as hiking contribute to a higher nutrient input and can accelerate eutrophication (Dynowski et al., 2019; Roşca et al., 2020; Senetra et al., 2020). These effects from hiking activities may also be amplified with climate change, as it is likely to extend the hiking season in addition to increase hiking tourism (Pröbstl-Haider et al., 2020).

These increasing pressures on mountain lakes require decision makers and managers to stronger focus on their management in order to satisfy the differing needs of lake visitors as well as to preserve such ecologically sensitive sites. Knowledge about preferences of visitors is central to define targeted management strategies that can effectively support tourism guidance and development as well as the integration of societal values and benefits into conservation policies (Plieninger et al., 2015; Schirpke et al., 2017; Willis, 2015). In particular, to prevent the occurrence of potential conflicts related to recreational use of mountain lakes, it is important to identify groups with individual characteristics and attitudes (Scolozzi et al., 2015). Although some studies have distinguished different recreational groups in various contexts (Fariás Torbidoni, 2011; Kienast et al., 2012; Komossa et al., 2018; Pröbstl-Haider et al., 2016; Schirpke et al., 2018), specific studies focusing on outdoor recreation at mountain lakes are still rare.

Therefore, this study aimed at assessing people's preferences related to a visit to natural mountain lakes in the European Alps. Specifically, we collected preferences related to (1) general aspects related to the visit, (2) recreational activities, (3) accessibility (hiking length, trail

preferences, transport), and (4) touristic facilities. Hypothesizing that different people have different preferences and behaviour due to their specific socio-cultural background (Schirpke et al., 2018; Martin et al., 2020), we also aimed at identifying recreational visitor groups. Using a questionnaire, we collected the data via an online survey targeting visitors to mountain lakes. We first describe the questionnaire and data analysis. Then, we present our results and discuss our findings with regard to management implications.

2. Materials and methods

2.1. Questionnaire

We prepared a questionnaire to collect different aspects related to a visit to mountain lakes including activities, accessibility (i.e., hiking time, trail preferences, transport), and the importance of touristic facilities (Table 1). At the beginning of the questionnaire, we described its purpose and provided a definition of mountain lakes as being “smaller-sized natural lakes, which are located at least 1000 m above sea level” to assure that respondents provided their answers regarding mountain lakes. Moreover, we asked the participants to indicate gender, age, their relation to the Alps, nature contact and lake affinity to analyze differences in preferences based on the socio-cultural background of the respondents. All questions were arranged in a digital questionnaire using Google Forms (<https://docs.google.com/forms/>).

2.2. Data collection and analysis

Before distributing the questionnaire, we performed a pre-test involving 17 participants. Based on their indications, we rephrased some questions or options. The final questionnaire was made available between July and December 2020 in three languages (English, German and Italian) and data were collected via an online survey using Google Forms (<https://docs.google.com/forms/>). We used non-probability sampling techniques (purposive sampling and voluntary response sampling) to include people with a certain interest in visiting mountain lakes. Therefore, the links to the questionnaires were posted in dedicated social media channels and distributed via newsletters or by sending invitations via email to members of Alpine hiking associations as well as associations of biologists, limnologists of different countries (Austria, Germany and Italy). Participation was voluntary and anonymous and all data were collected via a web survey without registering further information such as IP addresses than those provided by the respondents. All participants confirmed their voluntary participation and could withdraw at any time.

The responses of all completed questionnaires were registered in a database and analysed dependent on the type of question using SPSS Statistics (version 26, IBM, Armonk, NY, USA). For questions that required to choose one or more options (Q2, Q4-6), we counted how often each option was selected. For questions to which respondents were asked to indicate their preference based on a rating scale (Q1, Q3, Q7), we assigned numeric values to each option (0 = does not apply at all/does not really apply/not important at all, 1 = neutral/less important, 2 = rather applies/important, 3 = definitely applies/very important) and calculated mean preference scores.

To identify significant differences in behaviour and preferences based on distinct socio-cultural characteristics (Martin et al., 2020), we summarised the information provided by the respondents according to six socio-cultural variables (Table 2), omitting classes with low sample size. We then used Chi-Square test to assess the significance of differences between groups.

To identify recreational visitor groups, we applied hierarchical cluster analysis using Ward's linkage method with Euclidean distances. Input variables included the options of the questions (Q1-Q7, Table 1) as well as socio-cultural groups (Table 2). Based on the agglomeration coefficient, measuring heterogeneity as the distance at which clusters

Table 1

Questions related to different aspects of a visit to mountain lakes and response options.

Aspect	ID	Question	Options
Visit	Q1	You are planning an excursion and intend to spend time in nature. In relation, how appropriate or inappropriate are the following statements for you? (5-point rating scale from '1 = does not apply at all' to '5 = definitely applies')	Mountain lakes are the primary destination of my excursion When choosing my holiday or excursion destination, mountain lakes are a key criteria On my excursions, I pass by mountain lakes rather coincidentally I am planning visits to mountain lakes for particularly hot days
	Q2	With whom do you most enjoy visiting a mountain lake? (choose one or more options)	On your own Family Partner Friends Group Dog
Activities	Q3	I associate a visit to an alpine lake with the following activities. How inappropriate or appropriate are the following statements for you? (5-point rating scale from '1 = does not apply at all' to '5 = definitely applies')	Boating Bathing/swimming Having a picnic Camping Having a barbecue Fishing Relaxing Sunbathing Reading/listening to music Observing nature Boating
		What other activities do you associate with it? Please, briefly describe in a few words.	
Accessibility	Q4	How long would you like to be hiking? (choose one option)	Less than 1 h Up to 2 h Up to 4 h More than 4 h
	Q5	What type of trail do you prefer? (choose one option)	Paved roads Gravel roads Marked trails in gentle terrain Marked trails with steeper passages and narrow footpaths Unmarked trails in partially natural terrain with short rocky passages
Touristic facilities	Q6	Which of these types of transport would you prefer to shorten the hike to a mountain lake? (choose one or more options)	Own car Cable car Bicycle/mountain bike E-Bike Shuttle or hiking bus None at all Other
	Q7	How important are the following elements to you? (4-point rating scale from 'not important at all' to 'very important')	A restaurant/hotel right by the lake Hiking/walking trails along the lakeshore Footbridges or platforms, leading to the lake Resting or barbecue areas on the lakeshore Seating along the lakeshore Direct access to the lake Water sports facilities (e.g. boat rental) Children's playgrounds directly by the lake
		Which other elements are of particular importance to you?	

Table 2

Socio-cultural groups used to test for differences in behaviour and preferences as well as to group lakes users into different beneficiary groups.

Socio-cultural variables	Classification	Sample size (n)
Gender	Female	320
	Male	201
Age	16–39 years	253
	40+ years	256
Cultural background	German-speaking	294
	Italian-speaking	210
Relation to Alps	Low (visiting the Alps primarily as a tourist)	229
	Strong (living and grew up in the Alps)	199
Nature contact	High (spending time in nature at least several times a week)	335
	Low (spending time in nature less than several times a month)	187
Lake affinity	Low (<=3 visits to lakes a year)	255
	High (>=4 visits to lakes a year)	258

are formed, we decided for the two-cluster solution, dividing the respondents into two major groups (Figure A1).

3. Results

3.1. Respondent's characteristics

We collected 526 valid responses with different socio-cultural characteristics (Fig. 1). Respondents included more women and German-speaking people. Respondents visiting the Alps for touristic purposes built the greatest group with regard to their relation to the Alps, followed by residents that were also born in the Alps. More than half of the respondents reported that they usually spend time in nature at least several times a week or visit mountain lakes at least 4 times a year.

3.2. Visiting mountain lakes and associated activities

We evaluated the importance of mountain lakes for excursions (Q1, Table A1). Most respondents, in particular Italian-speaking respondents and people with high lake affinity, indicated that mountain lakes were the primary destination of their excursions (55.5%), while fewer respondents stated that they were planning visits to mountain lakes for particularly hot days (38.0%). More than a third of the respondents (35.4%) indicated that they passed by mountain lakes rather coincidentally on their excursions. Respondents indicated least often that mountain lakes were a key criteria for choosing their holiday or excursion destination (33.1%). With regard to the question with whom they most enjoy visiting mountain lakes (Q2, Table A1), the majority of respondents stated with friends (67.3%), their partner (63.1%) or their family (50.8%). Fewer respondents indicated a preference for visiting mountain lakes alone (43.2%), with a dog (19.4%) or with a group (16.5%).

In terms of activities associated with a visit to mountain lakes (Q3), respondents mostly indicated observing nature, relaxing and having a picnic, while least important were fishing, boating and having a barbecue (Fig. 2 and Table A1). In addition, respondents mentioned further activities such as taking a mental break/chilling out (34.3%), hiking (24.5%), photography (15.0%), meditating/practicing yoga (8.7%), creative activities such as writing and drawing (4.5%), socialising (3.5%), taking a physical break/refreshing (3.5%), and other types of sports such as mountain biking (2.8%).

Greatest differences between different socio-cultural groups were related to the cultural background (Table A1). Italian-speaking respondents valued most recreational activities at lakes higher than German-speaking respondents, except for bathing/swimming. Bathing/swimming was also more important for people <40 years, residents and those with high nature contact.

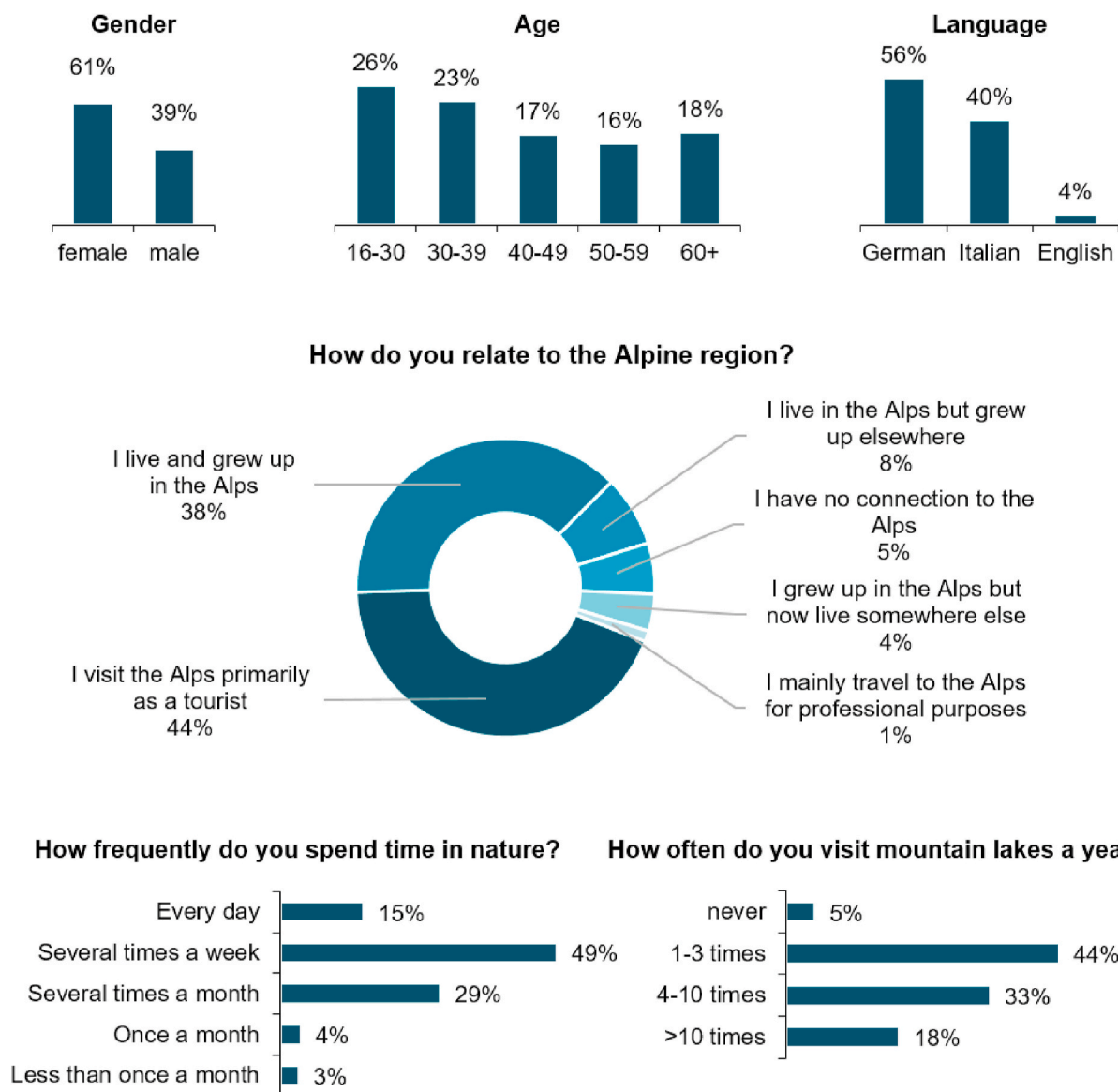


Fig. 1. Characteristics of the respondents (n = 526).

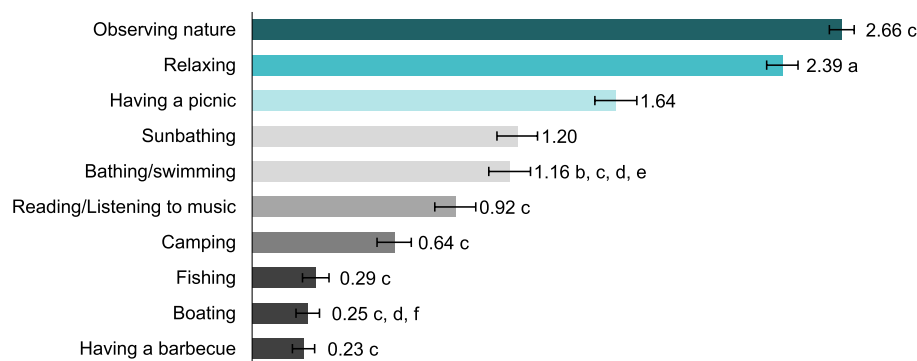


Fig. 2. Preference scores for activities associated with a visit to mountain lakes. Values with non-overlapping confidence intervals are significantly different from each other. Same colours of the bars indicate groups of activities for which the preference scores do not differ significantly from each other. Letters after the preference scores indicate statistically significant differences ($p < 0.01$) in preference scores between socio-cultural groups: (a) gender, (b) age, (c) cultural background, (d) relation to Alps, (e) nature contact and (f) lake affinity. Mean preference values for socio-cultural groups are reported in Table A1. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3.3. Preferences related to accessibility and touristic facilities

Longer hiking times to reach a mountain lake were generally preferred over shorter durations (Q4, Fig. 3 and Table A1). More than

half of the respondents preferred more than 4 h hiking (53.2%), while only 0.6% indicated less than 1 h. In terms of the type of trail (Q5), most respondents preferred marked trails with steeper passages or in gentle terrain, while gravel roads or paved roads were least preferred (Fig. 3

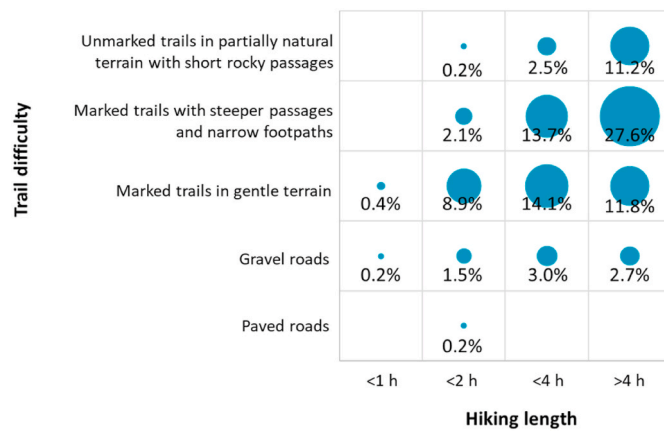


Fig. 3. Preferences related to different aspects of accessibility to mountain lakes based on 526 responses.

and Table A1). To shorten the way to a mountain lake (Q6), cable car was the most preferred type of transport (35.6%) followed by own car (27.0%), bicycle/mountain bike (25.3%), and shuttle or hiking bus (21.5%) (Table A1). However, most respondents (36.3%) stated that they preferred to reach a mountain lake without the support of means of transport.

With regard to touristic facilities at the lake (Q7), most respondents indicated that hiking/walking trails along the lakeshore as well as direct access to lake are important (Fig. 4 and Table A1). Important was also seating along the lakeshore, while playgrounds and water sports facilities were least important. Tourists and people with low nature contact or low lake affinity attributed higher importance to tourist facilities than residents and people with high nature contact or high lake affinity (Table A1).

3.4. Recreational visitor groups

Using the respondents' preferences with regard to the questions Q1-Q7 (see also Table 1) as well as their socio-cultural characteristics (Table 2), we applied cluster analysis to identify recreational visitor groups. Based on the respondents who provided full socio-cultural information ($n = 396$), we identified two major visitor groups to mountain lakes with diverging preferences (Fig. 5, Figure A1).

Group 1: nature-oriented visitors

This group (46.6%) generally prefers pristine and remote mountain lakes, i.e. lakes that are difficult to reach and that have no tourist facilities. However, lakes are in most cases not the primary destination of excursions and therefore recreational activities at the lake are less important, except for observing nature. This group is characterised by a higher share of residents and people with high nature contact and high

lake affinity.

3.4.1. Group 2: leisure-oriented visitors

This group (53.4%) favours mountain lakes that are developed for tourism, i.e., lakes that are easily accessible, also with the help of transportation such as cable cars, and that have touristic facilities. People of this group prefer visiting mountain lakes together with their family or in a group. Since mountain lakes are often the primary destination of excursions, especially on hot days, this group values recreational activities at the lake higher. Group 2 mostly includes tourists and people with lower nature contact and lower lake affinity.

4. Discussion

4.1. Outdoor recreation at mountain lakes

Our results confirm that mountain lakes are important destinations for outdoor recreation (Angradi et al., 2016; Hossu et al., 2019; Reynaud and Lanzanova, 2017; Sterner et al., 2020). Similar to other studies (Schirpke et al., 2021b; Scolozzi et al., 2015), our results indicate that passive recreation such as observing nature and relaxing is more important than active recreation. One explanation is the limited accessibility of mountain lakes. Mountain lakes are usually visited during a hiking excursion (Pröbstl-Haider et al., 2020), and their remoteness makes it difficult to bring equipment for recreational activities. Harsh climatic conditions are a further factor for limiting aquatic activities, as low water temperatures of mountain lakes are mostly unfavourable for bathing or swimming in contrast to lowland lakes (Hossu et al., 2019; Keeler et al., 2015; Pröbstl-Haider et al., 2020; Sterner et al., 2020). Accordingly, hiking/walking trails along the lakeshore as well as direct access to the lake were indicated as more important than water sport facilities such as boat rental. Other artificial infrastructure at the lake was less important, although facilities can support positive experiences and social gathering for some visitor groups (Vierikko and Yli-Pelkonen, 2019).

Some limitations related to the sample need to be taken into account for interpreting our results. Since we used non-probability sampling techniques, involving people supposed to be interested in visiting mountain lakes, our results do not sufficiently reflect the preferences of other types of visitors (only 5% of the respondents indicated that they never visit mountain lakes). Preferences related to hiking length and difficulty may also be influenced by the sample, as predominantly members of Alpine hiking associations received the link to the survey. Another issue is related to the use of an online survey preventing to ask for further explanations. Not being interviewed at a lake may also lead to discrepancies between the expressed preferences and actual behaviour. Nevertheless, the results from this exploratory study can be considered as representative for a large group of visitors to mountain lakes and applied for lakes that are mostly visited by German-speaking and Italian-speaking people. Future research should include respondents with other

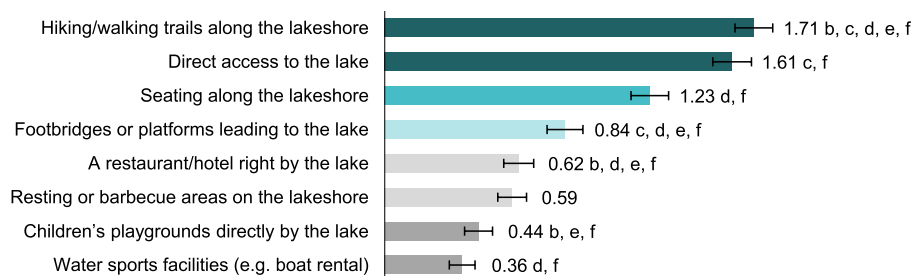


Fig. 4. Preference scores for touristic infrastructure at mountain lakes. Values with non-overlapping confidence intervals are significantly different from each other. Same colours of the bars indicate groups of activities for which the preference scores do not differ significantly from each other. Letters after the preference scores indicate statistically significant differences ($p < 0.01$) in preference scores between socio-cultural groups: (a) gender, (b) age, (c) cultural background, (d) relation to Alps, (e) nature contact and (f) lake affinity. Mean preference values for socio-cultural groups are reported in Table A1. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this

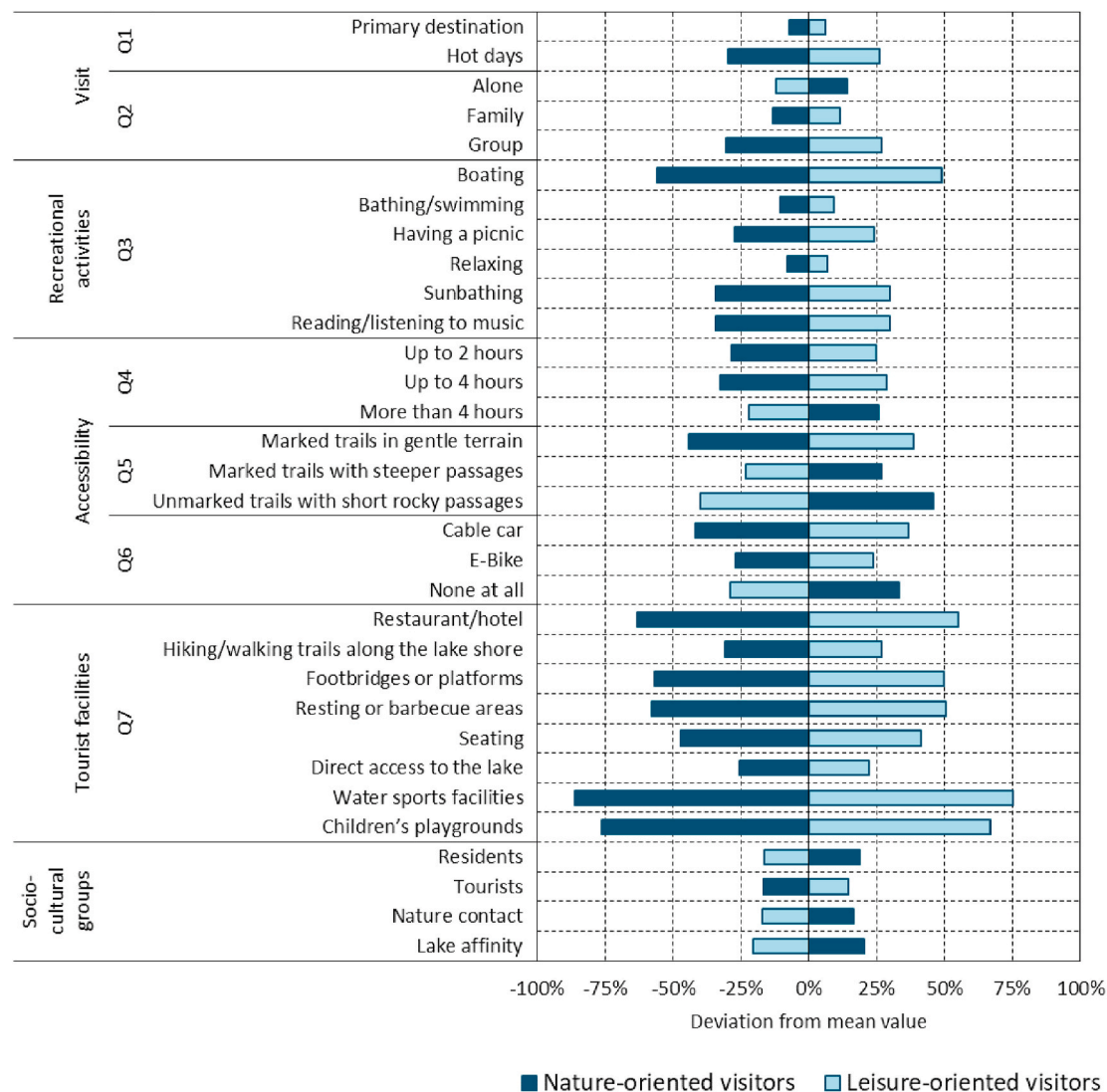


Fig. 5. Differences in preferences of the two recreational visitor groups for different aspects related to a visit to mountain lakes (Q1-Q7). Only significant differences ($p < 0.05$) are included.

cultural background to provide insights for lakes in the French or Slovenian Alps.

4.2. Recreational visitor groups and potential conflicts

We identified two major recreational visitor groups of mountain lakes, which reflect groups identified by other studies in hiking contexts (Farías Torbidoni, 2011; Pröbstl-Haider et al., 2016). With regard to their demands towards tourism infrastructure, nature-oriented visitors are comparable to nature-minded and sporting hikers (Farías Torbidoni, 2011) as well as experienced mountain tourists (Pröbstl-Haider et al., 2016), as this group prefers difficult trails and has low demands regarding tourist facilities. The group of leisure-oriented visitors matches general-purpose hikers (Farías Torbidoni, 2011) as well as casual mountain tourists (Pröbstl-Haider et al., 2016), seeking relaxation and attributing higher value to touristic facilities. Since we focused in our survey on respondents interested in hiking and mountaineering activities, further types of visitors or subgroups were not identified, including mountain bikers, mushroom pickers and culinary tourists (Schirpke et al., 2018; Scolozzi et al., 2015), anglers and water-based recreationists (Confer et al., 2005; Hunt, 2005), as well as spiritual, education and convenient recreationists (Komossa et al., 2018). A finer differentiation may therefore be useful when focusing on specific lakes

or a specific region. In particular, for lakes that are easily reachable, preferences of different groups of tourists, including those who are not interested in hiking, should be compared in future research.

Differing interests and preferences with regard to recreational use and touristic infrastructure may lead to behaviour that causes conflicts between different groups of recreational users (Confer et al., 2005; Kainzinger et al., 2015; Schirpke et al., 2020; Scolozzi et al., 2015). Our results suggest that the presence of touristic facilities at the lake may have negative impacts on recreational benefits of nature-oriented visitors, while its absence diminishes the benefits of leisure-oriented visitors. Direct conflicts may be caused by carrying out recreational activities at the lake such as having a barbecue, camping or listening to music, as such activities incite noise and pollution and lead to disturbance of other visitors (Confer et al., 2005; Scolozzi et al., 2015), in particular, of nature-oriented visitors. Nevertheless, the different preferences in accessibility may defuse such conflicts in regions with high lake abundance. In this case, leisure-oriented visitors are expected to concentrate on rather easily accessible and touristic lakes, while nature-oriented visitors continue the excursion to remote and less visited lakes.

4.3. Management implications

Considering the increasing demand for outdoor recreation in general (Buckley et al., 2015) and for cooler destinations due to climate change in specific (Pröbstl-Haider et al., 2020), hiking tourism in the European Alps is likely to intensify and visitation rates will also increase at mountain lakes. Therefore, management strategies are required to reduce the conflict potential between different visitor groups (Confer et al., 2005; Schirpke et al., 2020; Scolozzi et al., 2015) as well as to limit negative environmental impacts on mountain lakes and the surrounding landscape (Dynowski et al., 2019; Schmeller et al., 2018; Senetra et al., 2020).

To identify effective management strategies, it may be useful to differentiate between lakes that are easily accessible and remote lakes. Highest demand for management occurs for easily accessible lakes that have high visitation rates. Here, a targeted visitor management is necessary to reduce the conflict potential between user groups as well as negative environmental impacts (Marion et al., 2016). This can include a limitation in accessibility (e.g., no access with private car) or a distinction of areas for different purposes in case of larger lakes. For example, some areas may be equipped with tourist facilities such as playgrounds or restaurants to attract families, tourists or leisure-oriented visitors. Other areas instead may be left without facilities for people preferring more quiet places. For remote lakes, our results suggest limiting the construction of trails, transportation infrastructure and tourist facilities at the lake to support recreational opportunities for nature-oriented visitors and to avoid an increase in visitation rates. This can also help to preserve the high environmental quality of pristine lakes and limit negative impacts on sensitive ecosystems.

Our results also provide valuable information for trail management. In general, easy access (short walking time, gravel or paved roads, and availability of means of transport) was less preferred by the respondents. A widening or paving of hiking trails or the construction of infrastructure can consequently diminish the hiking experience associated with a visit to mountain lakes, although visitors with limited mobility such as families with small children or elderly people may have other requirements (Koppen et al., 2014). Moreover, easy access is closely related to higher visitation rates, which may lead to crowdedness and thus possibly reduce the recreational values of visitors who visited the place even without eased access (Confer et al., 2005; Roux et al., 2020; Zehrer and Raich, 2016), but further research is needed, as the level of perceived crowdedness can vary between different types of users due to different expectations (Luque-Gil et al., 2018).

In general, raising awareness for a responsible behaviour towards the lake ecosystem may contribute to reduce nutrient inputs from recreational activities within and around the lake (Dokulil, 2014; Senetra et al., 2020). As lakes smaller than 0.5 km² are currently not included in the Water Framework Directive of the European Union (EC, 2003), their water quality is usually not monitored, and lake owners do not need to implement specific management measures. However, fish feeding (Weckström et al., 2016) or livestock farming in the surroundings (Roşca et al., 2020; Van Colen et al., 2018) may require the introduction of regulations to sustain a high water quality, which is not only important for ecological reasons but also for aesthetic and recreational benefits (Angradi et al., 2018; Janssen et al., 2020; Keeler et al., 2015; Pröbstl-Haider et al., 2020; Schirpke, Scolozzi, & Tappeiner, 2021). Furthermore, diverse interests in the use of the lake, e.g., for livestock farming, irrigation, hydropower generation, fish production and angling, recreation, nature conservation (Fu et al., 2018; Reynaud and Lanzanova, 2017; Sterner et al., 2020), may cause further conflicts between different users or stakeholders (Schirpke et al., 2020). Management strategies therefore need to account for the specific local context, and decision makers should involve multiple groups of stakeholders and beneficiaries (Darvill and Lindo, 2016). To assure that also weaker or less represented groups are included, a qualitative stakeholder analysis,

e.g., SWOT analysis, can support the identification of all stakeholders and help to evaluate their influence (Schirpke et al., 2017). Fully exploring stakeholder opinions and values in a transparent process can help to better understand local conditions and dynamics and to anticipate or overcome conflicts that may emerge from disregarding diverse values and interests (Darvill and Lindo, 2016; Plieninger et al., 2015). Here, our findings provide novel perspectives about preferences of recreationists with regard to accessibility and touristic facilities as well as recreational values related to mountain lakes to decision-makers and tourism managers, which can be used to ensure opportunities for sustainable engagement with nature (Willis, 2015).

Our findings suggest that the behaviour of people, indicated by lake affinity and the level of nature contact, influenced in particular the preferences for touristic infrastructure at mountain lakes. This is reflected in the two recreational groups, but frequencies of lake visits or time in nature alone do not sufficiently reflect people's relationships with nature such as place identity, sense of belonging and nature connectedness (Martin et al., 2020). For example, tourists who are mostly represented by the group of leisure-oriented visitors may still hold high place attachment to a lake, even though visits are infrequently. A higher degree of attachment to a site, or to nature in general, may in turn influence environmental values and lead to a more responsible behaviour towards sensitive sites (Kaltenborn and Bjerke, 2002; Mackay and Schmitt, 2019; Martin et al., 2020). Decision-makers could encourage pro-environmental behaviours by providing more opportunities to engage with nature, such as ensuring accessibility to natural sites, maintaining the environmental quality and supporting interventions that aim at experiencing nature and that involve visitors by reflecting about emotions and values (Mackay and Schmitt, 2019; Passmore and Holder, 2017). In particular, leisure-oriented recreationists may benefit in many ways from such opportunities. These may not only increase physical and mental health benefits (McDougall et al., 2020; Völker and Kistemann, 2011), but taking on other perspectives can create new experiences and emotions and lead to deeper place attachment (Martin et al., 2020; Passmore and Holder, 2017). This may finally also avoid potential conflicts with nature-based recreationists.

5. Conclusions

This study advances the knowledge on outdoor recreation related to mountain lakes by providing insights into people's preferences regarding recreational activities, accessibility, and touristic facilities. Our findings suggest that the construction of new tourist facilities or transport infrastructure needs to be cautiously evaluated, while it is important to provide marked hiking trails. The distinction of the two visitor types can also help to develop targeted management strategies, differentiating easily accessible and remote lakes. Trail infrastructure and tourist facilities may be more developed for easily accessible lakes to attract leisure-oriented visitors in contrast to remote lakes that may be kept in a pristine state to support recreational benefits of nature-oriented visitors.

Most respondents of this study were German- or Italian-speaking and interested in hiking activities. Our findings are therefore mainly relevant for lakes that are located in the Austrian, German, Swiss and Italian Alps and visited by hikers. However, preferences of French and Slovenian people, international tourists and people who are less interested in hiking activities are still to be collected. Further research should focus on perceptions related to threats and conflicts at lakes with high visitation rates to improve visitor management and to preserve recreational ecosystem services as well as sensitive ecosystems.

CRedit authorship contribution statement

Uta Schirpke: Conceptualization, Methodology, Formal analysis, Data curation, Writing – original draft. **Rocco Scolozzi:** Formal analysis, Writing – review & editing. **Alexander Kiessling:** Methodology,

Resources, Writing – review & editing. **Ulrike Tappeiner:** Methodology, Writing – review & editing.

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Appendix

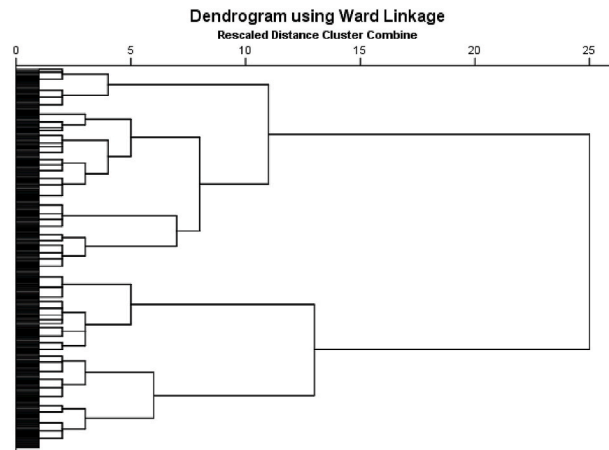


Fig. A1. Dendrogram of the cluster analysis for grouping respondents based on their preferences and socio-cultural characteristics into recreational visitor groups.

Table A1

Mean preferences (in descending order) related to the visit (Q1 and Q2), activities (Q3), accessibility (Q4 to Q6) and touristic facilities (Q7), and distinguished by socio-cultural groups. Statistically significant differences between groups are indicated by * ($p < 0.01$) and *** ($p < 0.001$).

Option	Total	Gender		Age		Cultural background		Relation to Alps		Nature contact		Lake affinity	
		Female	Male	<40	40+	German	Italian	Residents	Tourists	Low	High	Low	High
	N =	N =	N =	N =	N =	N =	N =	N = 199	N =	N =	N =	N =	N =
	526	320	201	253	256	294	210		229	187	335	255	268
Q1 Primary destination	1.631*	1.731*	1.483*	1.502*	1.785	1.303**	2.095**	1.688	1.672	1.684	1.603	1.376**	1.869**
Hot days	1.165	1.206	1.114	1.099	1.230	1.235	1.048	1.231	1.157	1.059	1.224	1.075	1.257
Key criteria	1.141*	1.238*	0.990**	0.937**	1.336	0.871**	1.548**	1.085	1.162	1.209	1.096	0.937**	1.328**
Rather coincidentally	1.105	1.106	1.104	1.134	1.105	1.337**	0.795**	1.065	1.201	1.021	1.155	1.322**	0.907**
Q2 Friends	0.673	0.669	0.682	0.810**	0.547**	0.684	0.638	0.628	0.690	0.690	0.666	0.667	0.675
Partner	0.631	0.594	0.692	0.700*	0.570*	0.687*	0.552*	0.638	0.646	0.631	0.636	0.631	0.634
Family	0.508	0.525	0.483	0.561	0.461	0.541	0.471	0.538	0.507	0.503	0.507	0.537	0.478
Alone	0.432	0.397	0.483	0.490	0.379	0.435	0.438	0.518**	0.358**	0.374	0.463	0.345**	0.519**
Dog	0.194	0.241**	0.114**	0.213	0.176	0.177	0.229	0.201	0.192	0.134*	0.227*	0.176	0.209
Group	0.165	0.141	0.199	0.186	0.148	0.136*	0.205*	0.141	0.192	0.176	0.158	0.165	0.168
Q3 Observing nature	2.660	2.709	2.587	2.644	2.699	2.568**	2.790**	2.724	2.677	2.674	2.654	2.604	2.709
Relaxing	2.392	2.469*	2.269*	2.470	2.313	2.446	2.276	2.442	2.367	2.358	2.406	2.408	2.369
Having a picnic	1.641	1.681	1.572	1.696	1.586	1.616	1.771	1.653	1.664	1.733	1.585	1.635	1.649
Sunbathing	1.196	1.222	1.164	1.194	1.223	1.126	1.367	1.171	1.197	1.241	1.173	1.094	1.291
Bathing/ swimming	1.162	1.166	1.159	1.328*	1.020*	1.391**	0.876**	1.296*	1.004*	0.973*	1.269*	1.043	1.280
Reading/ listening to music	0.916	0.969	0.811	0.996	0.848	0.673**	1.262**	0.905	0.926	1.016	0.854	0.961	0.862
Camping	0.641	0.613	0.687	0.723	0.578	0.497**	0.843**	0.673	0.642	0.674	0.624	0.557	0.720
Fishing	0.287	0.241	0.363	0.241	0.344	0.112**	0.557**	0.332	0.245	0.337	0.251	0.212	0.354
Boating	0.251	0.284	0.189	0.221	0.285	0.170*	0.343*	0.161*	0.319*	0.289	0.224	0.345**	0.160**
Having a barbecue	0.232	0.238	0.229	0.269	0.203	0.167*	0.333*	0.286	0.201	0.262	0.215	0.216	0.246
Q4 More than 4 h	0.532	0.516	0.552	0.581	0.484	0.466**	0.638**	0.553	0.541	0.487	0.561	0.412**	0.649**
Up to 4 h	0.333	0.350	0.308	0.300	0.367	0.361	0.271	0.337	0.328	0.358	0.313	0.365	0.299
Up to 2 h	0.129	0.128	0.134	0.119	0.137	0.163	0.090	0.106	0.127	0.150	0.119	0.216**	0.049**

(continued on next page)

Table A1 (continued)

	Option	Total	Gender		Age		Cultural background		Relation to Alps		Nature contact		Lake affinity	
			Female	Male	<40	40+	German	Italian	Residents	Tourists	Low	High	Low	High
		N = 526	N = 320	N = 201	N = 253	N = 256	N = 294	N = 210	N = 199	N = 229	N = 187	N = 335	N = 255	N = 268
Q5	Less than 1 h	0.006	0.006	0.005	0.000	0.012	0.010	0.000	0.005	0.004	0.005	0.006	0.008	0.004
	Marked trails with steeper passages	0.433	0.453	0.413	0.474	0.410	0.456	0.410	0.528*	0.380*	0.374	0.466	0.337**	0.526**
	Marked trails in gentle terrain	0.352	0.381	0.299	0.265**	0.430**	0.327	0.386	0.291	0.406	0.455**	0.296**	0.486**	0.224**
	Unmarked trails with short rocky passages	0.139	0.078**	0.229**	0.190**	0.078**	0.167*	0.086*	0.151	0.118	0.096	0.161	0.075**	0.198**
Q6	Gravel roads	0.074	0.084	0.060	0.067	0.082	0.048*	0.119*	0.030*	0.096*	0.070	0.078	0.098	0.052
	Paved roads	0.002	0.003	0.000	0.004	0.000	0.003	0.000	0.000	0.000	0.005	0.000	0.004	0.000
	None at all	0.363	0.356	0.378	0.395	0.332	0.313	0.424	0.367	0.358	0.385	0.346	0.294*	0.425*
	Cable car	0.356	0.406**	0.264**	0.320	0.383	0.361	0.362	0.327	0.393	0.380	0.343	0.420*	0.295*
Q7	Own car	0.270	0.256	0.284	0.281	0.262	0.313	0.219	0.307	0.249	0.193*	0.316*	0.216*	0.325*
	Bicycle/ mountain bike	0.253	0.191**	0.353**	0.344**	0.164**	0.337**	0.129**	0.312*	0.183*	0.203	0.284	0.243	0.265
	Shuttle or hiking bus	0.215	0.206	0.224	0.198	0.238	0.265*	0.148*	0.211	0.214	0.214	0.215	0.212	0.220
	E-Bike	0.120	0.103	0.144	0.111	0.133	0.105	0.148	0.171*	0.087*	0.107	0.128	0.110	0.131
	Hiking/walking trails	1.713	1.794	1.577	1.589*	1.871*	1.476**	2.005**	1.513**	1.913**	1.925**	1.591**	1.902**	1.537**
	Direct access to the lake	1.612	1.609	1.597	1.605	1.617	1.806**	1.376**	1.513	1.646	1.658	1.579	1.792**	1.444**
	Seating along the lakeshore	1.230	1.281	1.149	1.194	1.281	1.248	1.176	1.065*	1.336*	1.348	1.152	1.463	1.011
	Footbridges or platforms	0.837	0.906	0.716	0.763	0.883	0.65**	1.071**	0.658**	1.000**	1.091**	0.696**	1.043**	0.649**
	A restaurant/ hotel	0.622	0.681	0.532	0.498**	0.734**	0.554*	0.729*	0.452**	0.747**	0.791**	0.531**	0.800**	0.459**
	Resting or barbecue areas	0.591	0.591	0.582	0.589	0.590	0.605	0.576	0.573	0.629	0.684	0.531	0.682	0.507
	Children's playgrounds	0.435	0.425	0.453	0.336*	0.516*	0.412	0.481	0.367	0.502	0.551*	0.367*	0.533*	0.347*
	Water sports facilities	0.359	0.406	0.289	0.308	0.398	0.333	0.395	0.256*	0.450*	0.455*	0.310*	0.471**	0.257**

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